(6) $A = C = F$ and $B = D = E = 0$, the general equation yields an empty set $x^2 + y^2 + 1 = 0$, as there is no real solution.

(7) $A \neq 0$ or $C \neq 0$ and others are zeros, the general equation yield coordinate axes.

(8) $A = -C$ and rests are zero, the general equation yields a pair of lines $x^2 - y^2 = 0$.

Example 5.28

Identify the type of the conic for the following equations:

(1) $16y^2 = -4x^2 + 64$ (2) $x^2 + y^2 = -4x - y + 4$

(3) $x^2 - 2y = x + 3$ (4) $4x^2 - 9y^2 - 16x + 18y - 29 = 0$

Solution

Q.no.	Equation	condition	Type of the conic
1	$16y^2 = -4x^2 + 64$	3	Ellipse
2	$x^2 + y^2 = -4x - y + 4$	1	Circle
3	$x^2 - 2y = x + 3$	2	parabola
4	$4x^2 - 9y^2 - 16x + 18y - 29 = 0$	4	Hyperbola

EXERCISE 5.3

Identify the type of conic section for each of the equations.

1. $2x^2 - y^2 = 7$ 2. $3x^2 + 3y^2 - 4x + 3y + 10 = 0$ 3. $3x^2 + 2y^2 = 14$

4. $x^2 + y^2 + x - y = 0$ 5. $11x^2 - 25y^2 - 44x + 50y - 256 = 0$ 6. $y^2 + 4x + 3y + 4 = 0$

5.5 Parametric form of Conics

5.5.1 Parametric equations

Suppose $f(t)$ and $g(t)$ are functions of $'t'$. Then the equations $x = f(t)$ and $y = g(t)$ together describe a curve in the plane . In general $'t'$ is simply an arbitrary variable, called in this case a **parameter**, and this method of specifying a curve is known as **parametric equations**. One important interpretation of $'t'$ is time . In this interpretation,the equations $x = f(t)$ and $y = g(t)$ give the position of an object at time $'t'$.

So a parametric equation simply has a third variable, expressing x and y in terms of that third variable as a parameter . A parameter does not always have to be $'t'$. Using $'t'$ is more standard but one can use any other variable.

(i) Parametric form of the circle $x^2 + y^2 = a^2$

Let $P(x, y)$ be any point on the circle $x^2 + y^2 = a^2$.

Join OP and let it make an angle θ with x-axis.

Draw PM perpendicular to x-axis. From triangle OPM,

$x = OM = a\cos\theta$

$y = MP = a\sin\theta$

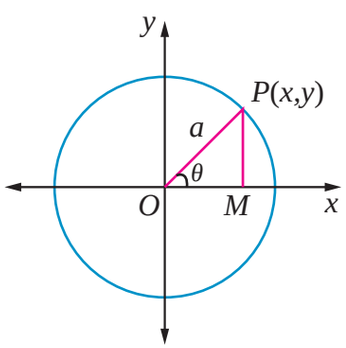


Fig. 5.44

Thus the coordinates of any point on the given circle are $(a\cos\theta, a\sin\theta)$ and

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